PF LAB 2

Task 1:

Run these Program and observe thier output.

1. Character Variable.

```
#include<iostream>
using namespace std;
int main()
{
char letter;
letter = 'A';
cout << letter << '\n';
letter = 'B';
cout << letter << '\n';
cin >> letter;
return 0;
}
```

Output:

A
B

2.

```
#include<iostream>
using namespace std;
int main()
{
char letter;
letter = 65;
cout << letter << '\n';
letter = 66;
cout << letter << '\n';
cin >> letter;
return 0;
}
```

Output:

A
B

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Task 2:

Take two integers as input from the user apply arthmetic operations on them (+, -, *, /) as print on screen.

```
#include<iostream>
using namespace std;
  int main()
  {
  int a, b;
cout<<"Enter two integers: ";
  cin>>a>>b;
cout<<" Addition = "<<a+b<<endl;
cout<<" Subtraction = "<<a-b<<endl;
cout<<" Multiplication = "<<a*b<<endl;
if(b!=0)
  cout<<" bivision = "<<(double)a/b<<endl;
  else
  cout<<"Division is not Possible."<<endl;
  return 0;
  }
```

Output: Enter two integers: 10 5

Addition = 15
Substraction = 5
Multiplication = 50
Division = 2

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Task 3:

Write the output.

```
#include "stdafx.h"
#include<iostream>
using namespace std;
int main()
{
    int val1;
    int val2;
    int val3;
cout << "please enter your 3 numbers!";
cin>> val1 >> val2 >> val3;
cout<<val1<< "\n"<<val2<<"\n"<<
val3 << "\n";
cout << val3 << "\n"<<val2<<"\n"<<
 val1<<"\n";
system("PAUSE");
return 0;
}
```

Output: Please enter your 3 number3 = 10, 20, 30

10
20
30

30
20
10

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Task 4:

Write a C++ Program to calculate the distance between the two points.

Note: x_1, y_1, x_2, y_2 are all double values.

```
#include<iostream>
using namespace std;
int main()
{
double x1, y1, x2, y2, distance;
cout<<" Enter x1 and y1; ";
cin>>x1>> y1;
cout<<" Enter x2 and y2 : ";
  cin>> x2 >> y2;
  distance = sqrt(pow (x2-x1,)+pow(y2-y1,2));
cout<<"Distance between two points = "<<
  distance <<endl;
  return 0;
  }
```

Output:

```
Enter x1 and y1 = 1 2
Ente x2 and y2 = 4 6
Distance between two points = 5
```

Task 5:

1- Calculate the Area of a circle. (area = PI * r2).

```
#include<iostream>
using namespace std;
int main()
{

double r, area;
const double PI = 3.14159;
cout<<"Enter radius:";
cin>>r;
area = PI * r * r;
cout<<" Area of circle = "<<area<<endl;
return 0;
}
```

Output:

```
Enter radius = 5
Area of circle = 78.5397
```

2. Calculate Area of Rectangle.

```
#include<iostream>
using namespace std;
int main()
{
double length, wiath, area;
cout<<"Enter length:";
cin >> length;
cout<<" Enter width:";
cin >> width;
area = length * width;
cout<<"Area of Rectangle=" <<area<<
    return 0;
    }
```

Output:

```
Enter length = 10
Enter width = 5
Area of Rectangle: 50
```

3. Calculate The Area of a Triangle.

Output=

Enter base = 10
Enter heigh = 6
Area of triangle = 30

```
#include<iostream>
using namespace std;
int main()
{
double base, height, area;
cout<<"Enter base:";
cin base;
cout<<"Enter height:";
cin >> height;
area = base * height * 0.5;
cout<<"Area of Triangle = "<<area<<endl;
retur 0;
}
```

Task 6: A customer in a store is purchasing five items The prices of the five items are:

price of item1 = $12.95
Price of item2 = $24.95
price of item3 = $6.95
price of item4 = $14.95
Price of item5 = $3.95

```
#include<iostream>
using namespace std;
int main()
{
double item1 = $12.95
double item2 = $24.95
double item3 = $6.95
double item4 = $14.95
double item5 = $3.95
```

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```
double subtotal, tax, total;
Subtotal = item1+item2+item3+item4+item5;
tax = Subtotal * 0.06;
total = Subtotal + tax;
cout << fixed << setprecision(2);
cout << " price of item 1 = $ " << item1 << endl;
cout << "Price of item 2 = $" << item2 << endl;
cout << "price of item 3 = $" << item3 << endl;
cout << "price of item 4 = $" << item4 << endl;
cout << "price of item 5 = $" << item5 << endl;

cout << "subtotal = $ " << subtotal << endl;
cout << "sales Tax = $" << tax << endl;
cout << "Total = $" << total << endl;
return 0;
}
```

output=

```
Price of item = $ 12.95
Price of item = $ 24.95
Price of item = $ 6.95
Price of item = $ 14.95
Price of item = $ 3.95

Subtotal = $ 63.75
Sale tax = $ 3.83
Total = $ 67.58
```

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Coding C++

Auto saved at 08:33:10

RUN MENU

```cpp
#include <iostream>
using namespace std;

int main()
{
    char letter;
    letter = 'A';
    cout << letter << '\n';

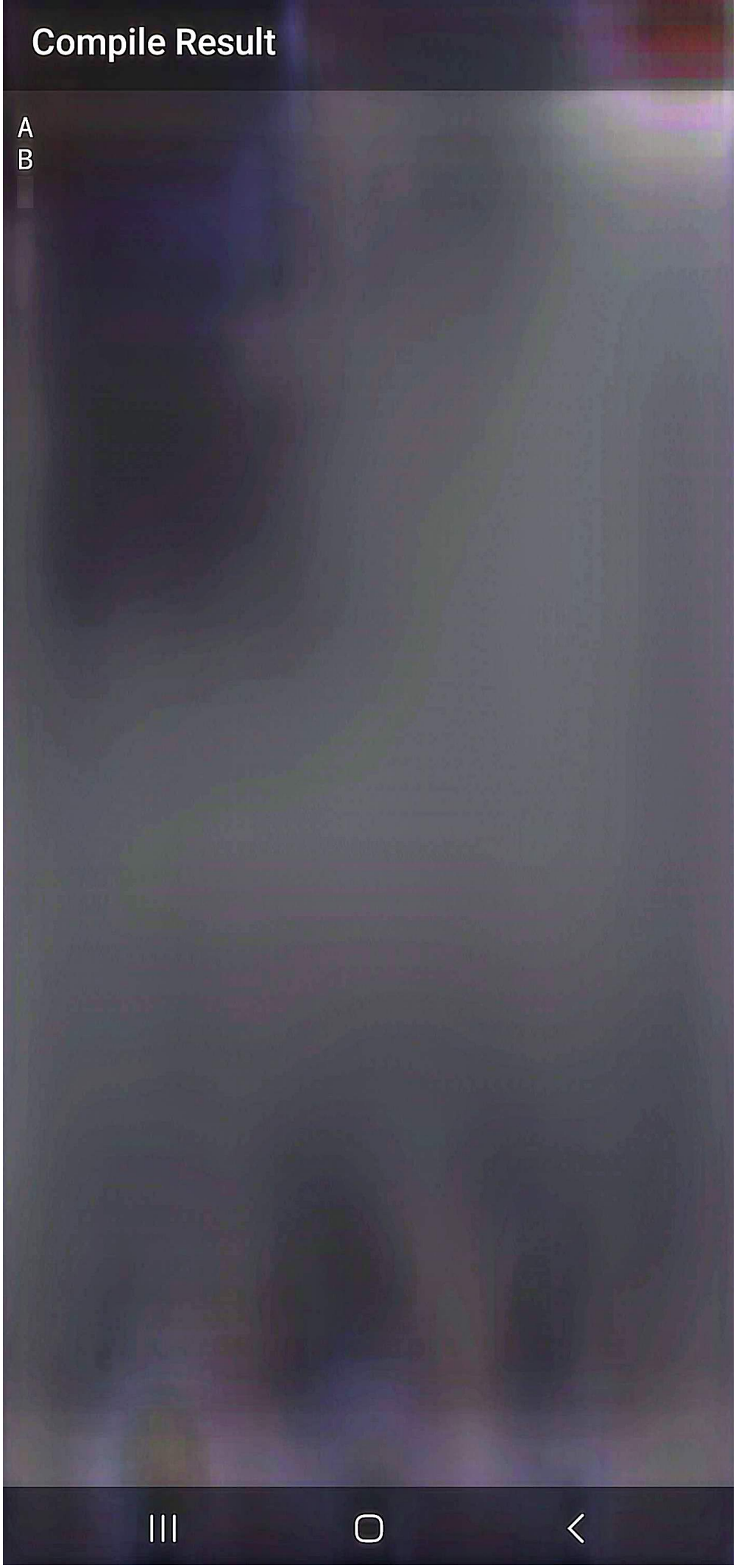
    letter = 'B';
    cout << letter << '\n';

    cin >> letter;

    return 0;
}
```

Tab {} “” ; = \ & ,

CamScanner



CamScanner

Coding C++

Auto saved at 08:36:48

RUN MENU

```cpp
#include <iostream>
using namespace std;

int main()
{
    char letter;
    letter = 65;
    cout << letter << '\n';

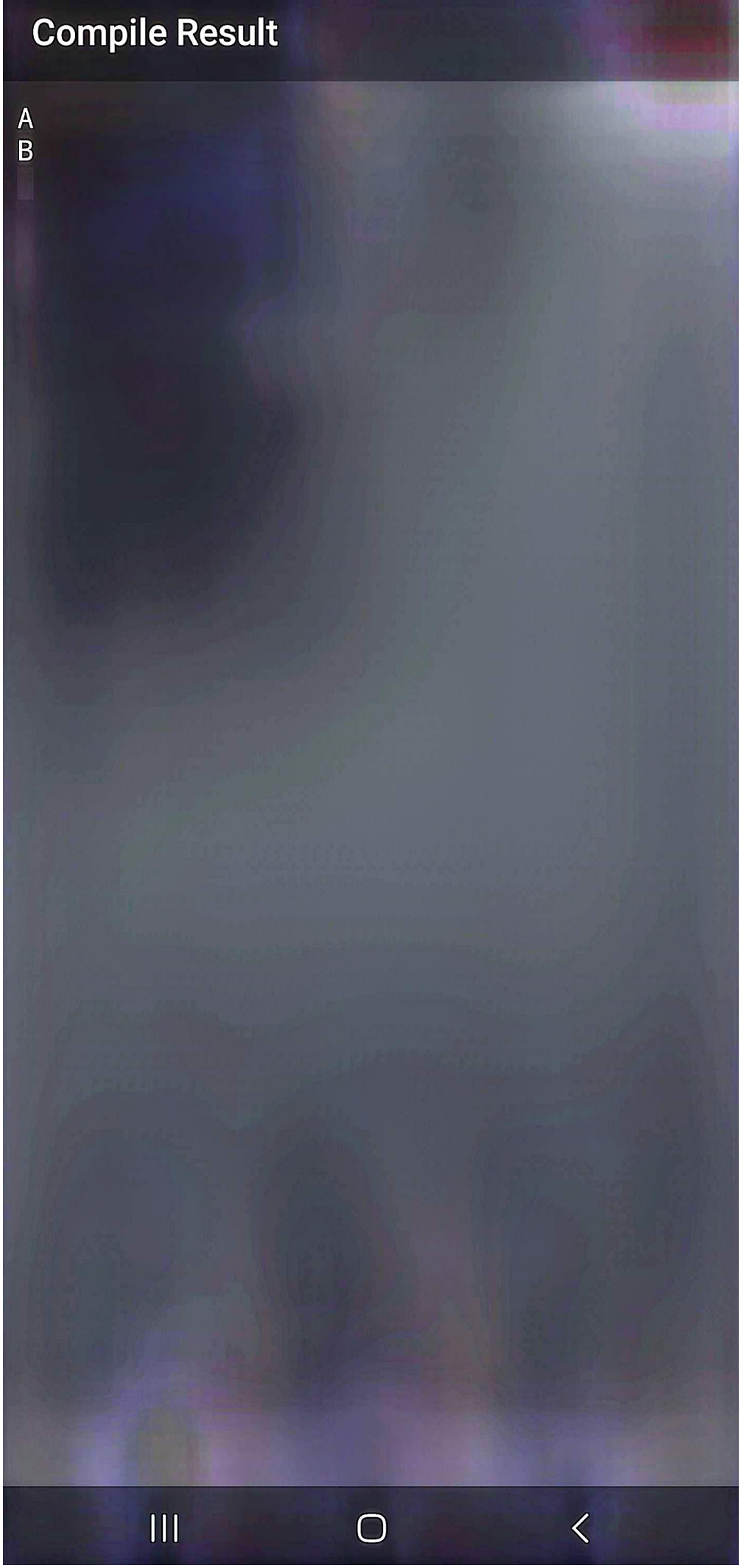
    letter = 66;
    cout << letter << '\n';

    cin >> letter;

    return 0;
}
```

Tab {} “” ; = \ & ,

CamScanner



CamScanner

Coding C++

Auto saved at 08:37:24

RUN MENU

```cpp
#include <iostream>
using namespace std;

int main()
{
    int letter;
    letter = 'A';
    cout << letter << '\n';

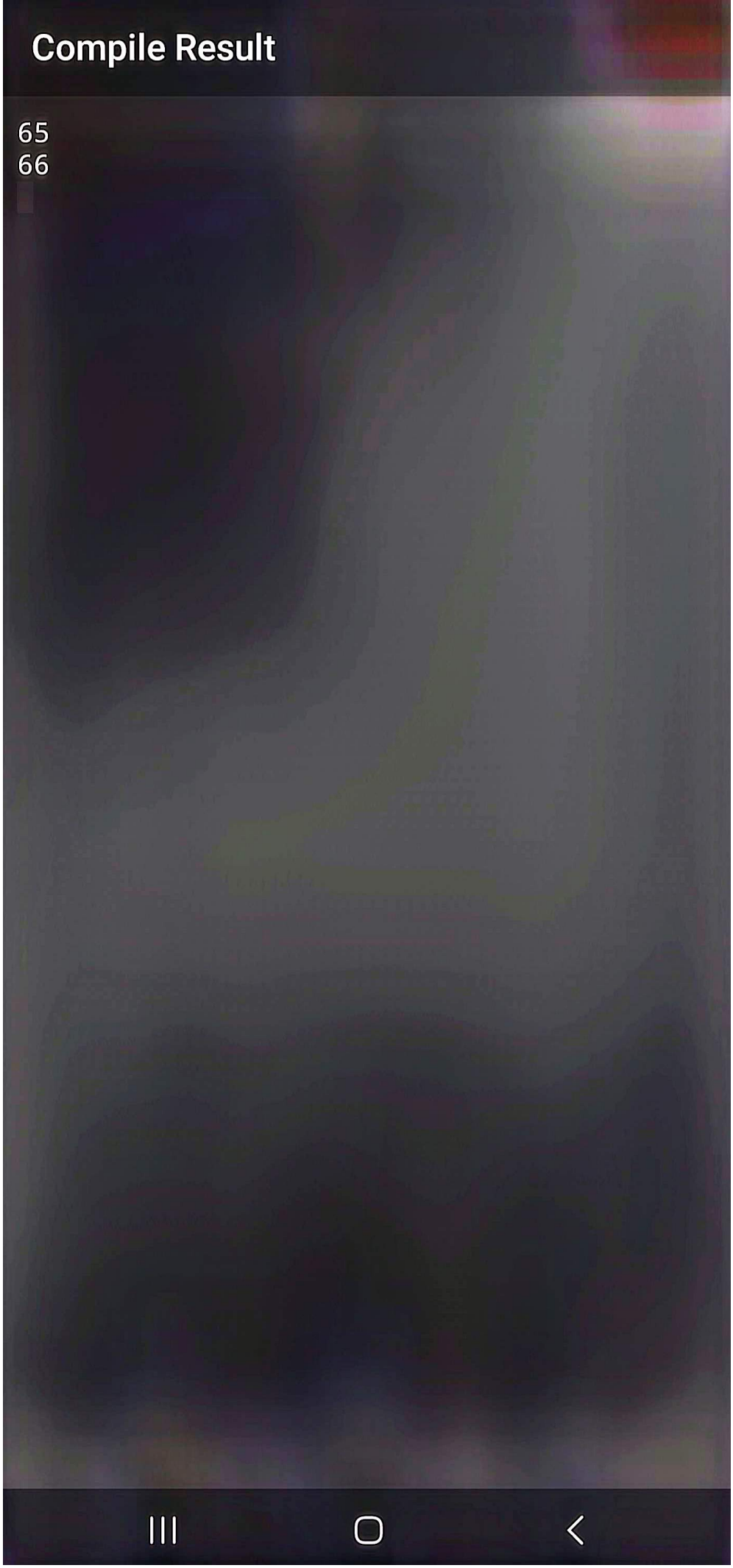
    letter = 'B';
    cout << letter << '\n';

    cin >> letter;

    return 0;
}
```

Tab {} “” ; = \ & ,

CamScanner



CamScanner

Coding C++

Auto saved at 08:38:02

RUN MENU

```cpp
#include <iostream>
using namespace std;

int main()
{
    int a, b;

    cout << "Enter two integers: ";
    cin >> a >> b;

    cout << "Addition = " << a + b << endl;
    cout << "Subtraction = " << a - b << endl;
    cout << "Multiplication = " << a * b << endl;

    if (b != 0)
        cout << "Division = " << (double)a / b << endl;
    else
        cout << "Division is not possible." << endl;

    return 0;
}
```

Tab {} “” ; = \ & ,

CamScanner

Compile Result

```
Enter two integers: 10 5
Addition = 15
Subtraction = 5
Multiplication = 50
Division = 2

[Process completed - press Enter]
```

CamScanner

Coding C++

Auto saved at 08:58:50

RUN MENU

```cpp
#include <iostream>
using namespace std;

int main()
{
    int val1;
    int val2;
    int val3;

    cout << "Please enter your 3 numbers: ";
    cin >> val1 >> val2 >> val3;

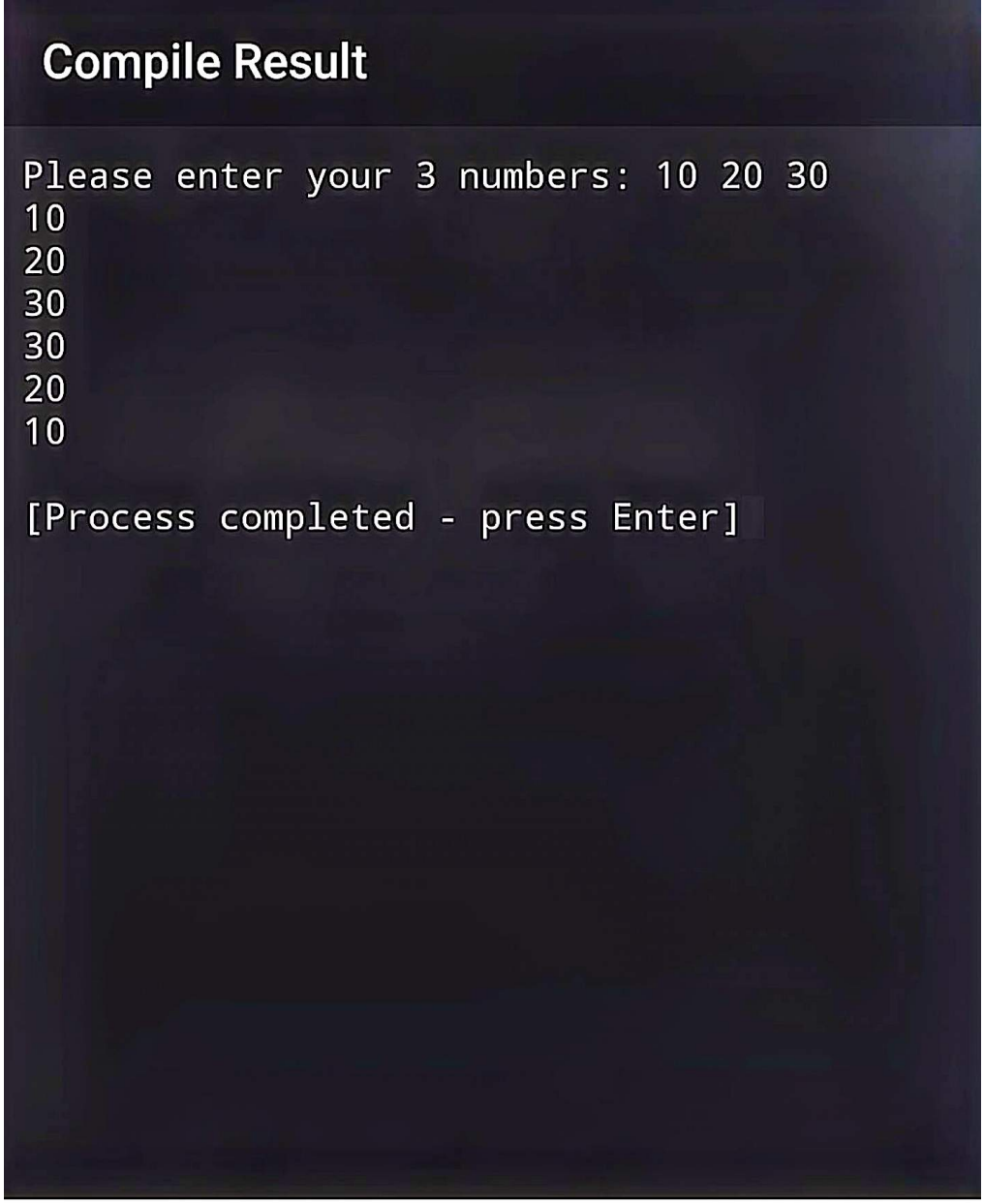
    cout << val1 << "\n" << val2 << "\n" << val3 << "\n";

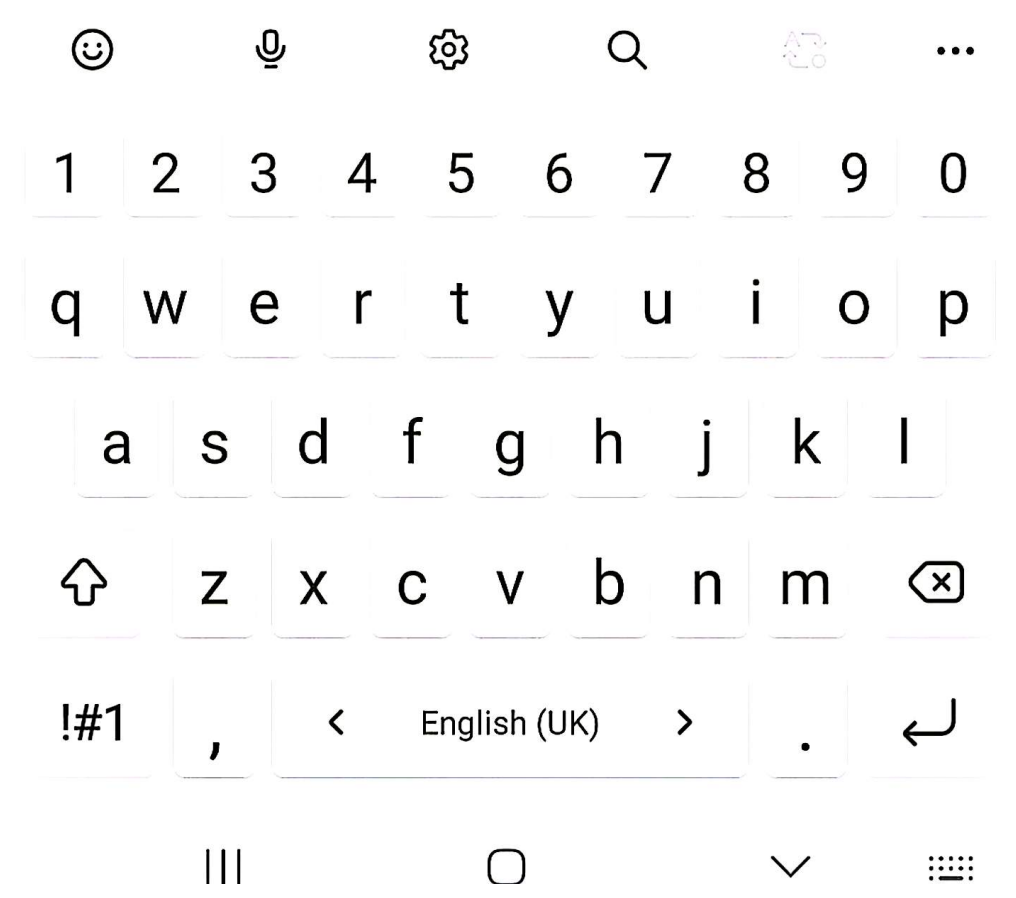
    cout << val3 << "\n" << val2 << "\n" << val1 << "\n";

    return 0;
}
```

Tab {} “” ; = \ & ,

CamScanner


Compile Result
Please enter your 3 numbers: 10 20 30
10
20
30
30
20
10
[Process completed - press Enter]


1 2 3 4 5 6 7 8 9 0
q w e r t y u i o p
a s d f g h j k l
z x c v b n m
!#1
English (UK)

CamScanner

Coding C++

Auto saved at 09:05:19

RUN MENU

```cpp
#include <iostream>
#include <cmath>
using namespace std;

int main()
{
    double x1, y1, x2, y2, distance;

    cout << "Enter x1 and y1: ";
    cin >> x1 >> y1;

    cout << "Enter x2 and y2: ";
    cin >> x2 >> y2;

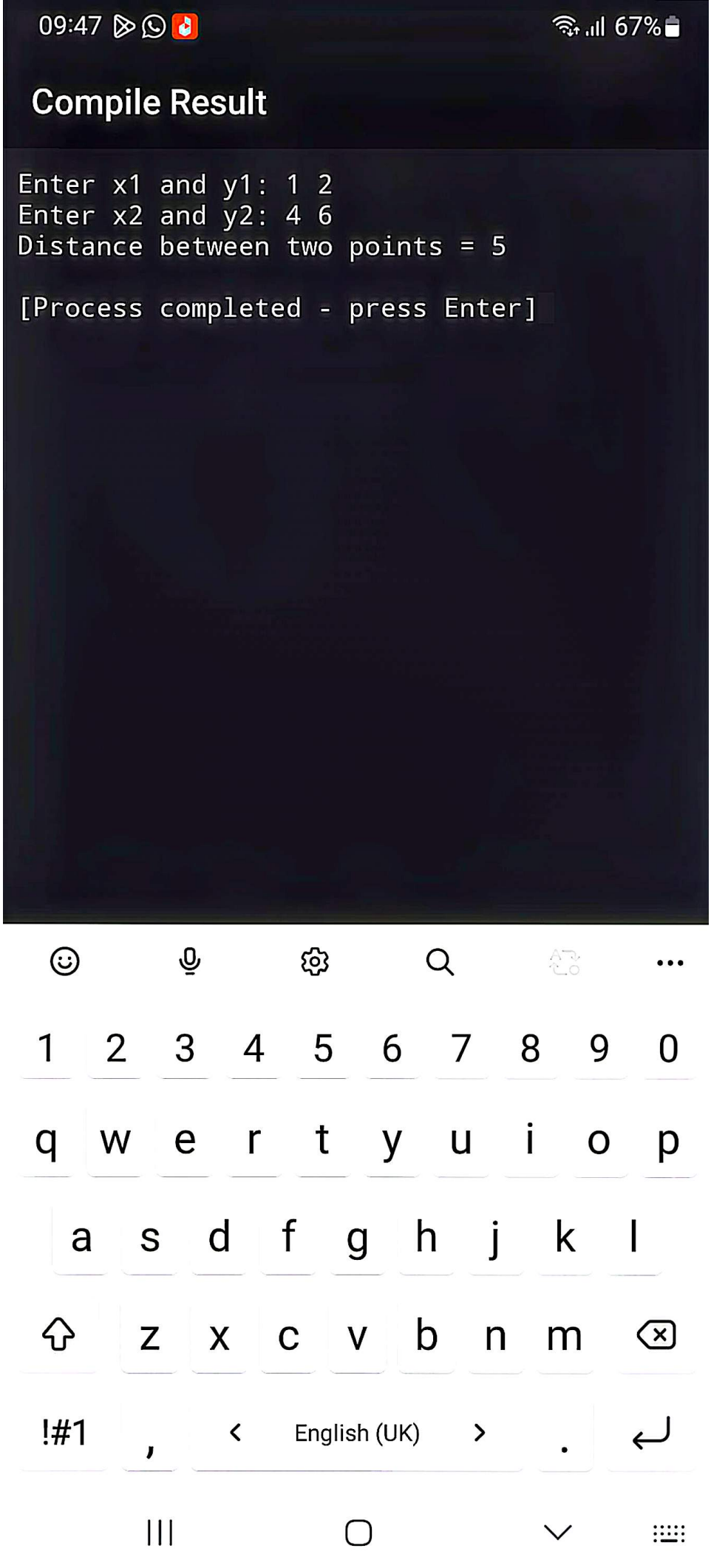
    distance = sqrt(pow(x2 - x1, 2) + pow(y2 - y1, 2));

    cout << "Distance between two points = "
         << distance << endl;

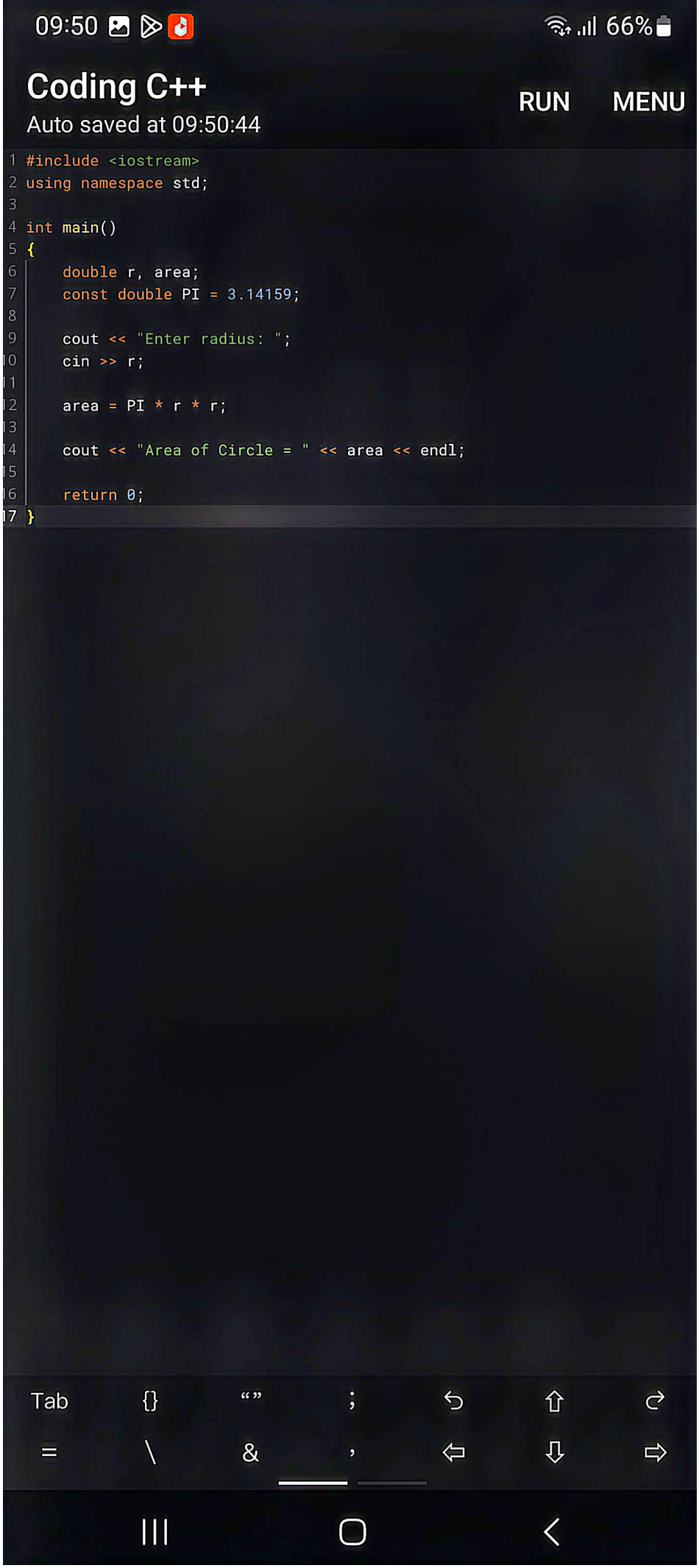
    return 0;
}
```

Tab {} “” ; = \ & ,

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09:47
67%
Compile Result
Enter x1 and y1: 1 2
Enter x2 and y2: 4 6
Distance between two points = 5
[Process completed - press Enter]
1 2 3 4 5 6 7 8 9 0
q w e r t y u i o p
a s d f g h j k l
z x c v b n m
!#1
English (UK)

CamScanner


09:50
66%
Coding C++
Auto saved at 09:50:44
RUN
MENU
#include <iostream>
using namespace std;
int main()
{
double r, area;
const double PI = 3.14159;
cout << "Enter radius: ";
cin >> r;
area = PI * r * r;
cout << "Area of Circle = " << area << endl;
return 0;
}
Tab
{}
“”
;
=
\
&
,

CamScanner

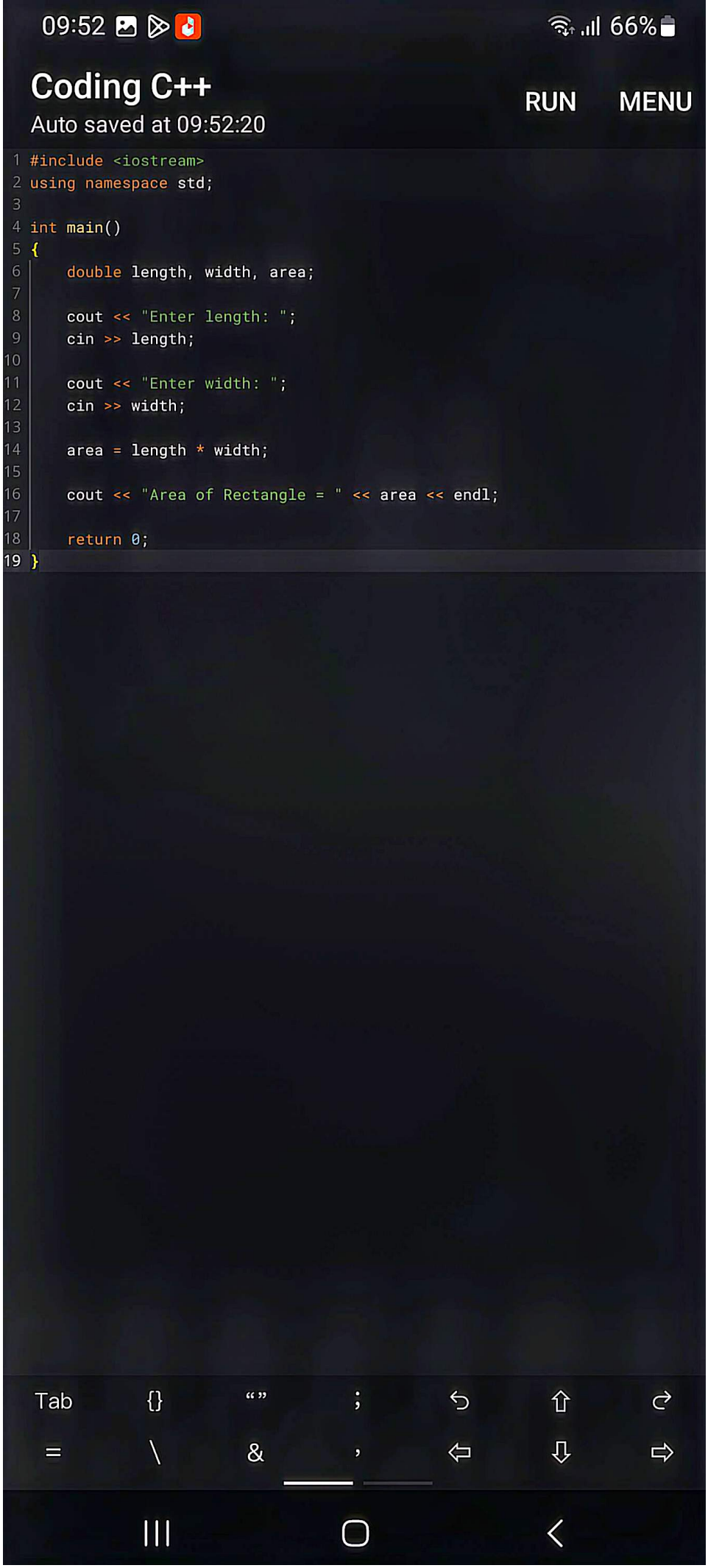
09:51 66%

Compile Result

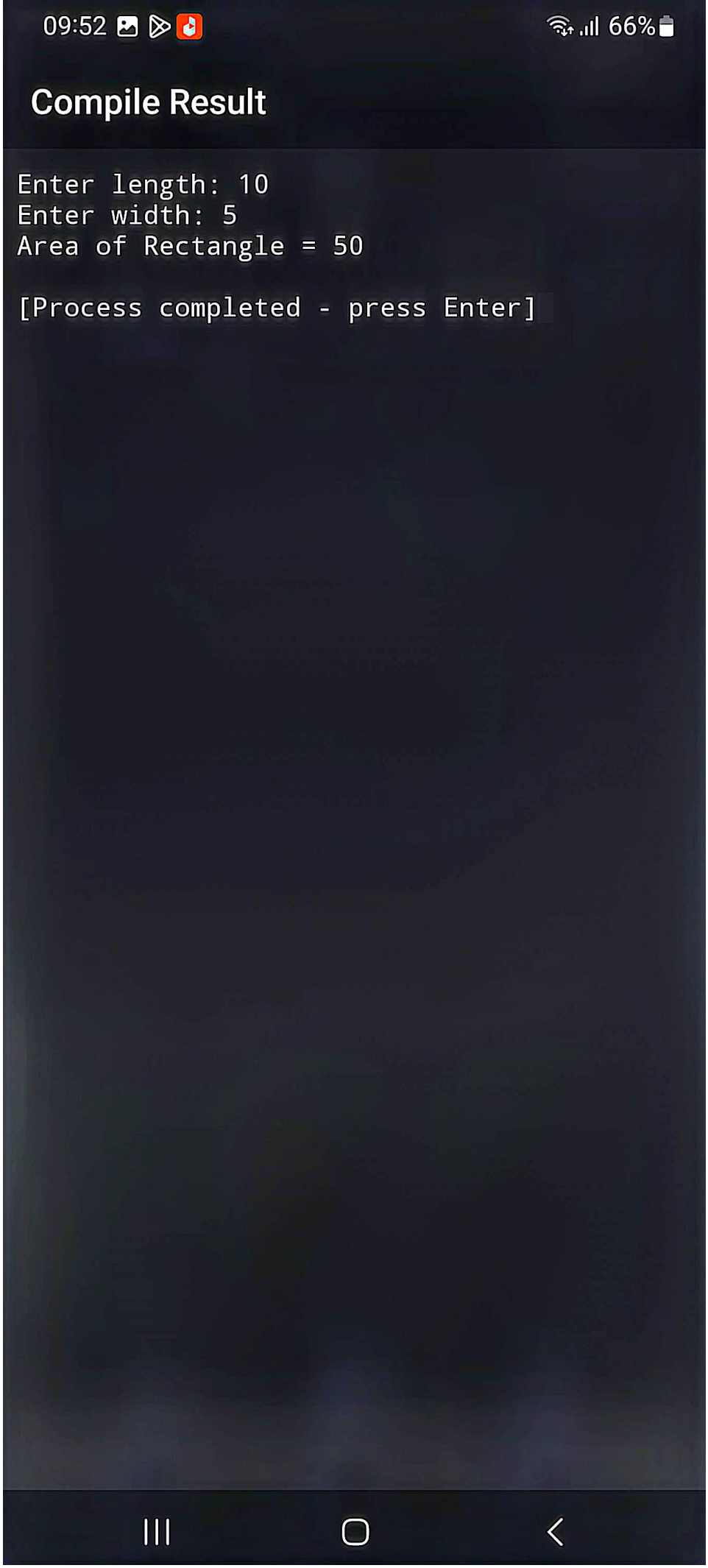
```
Enter radius: 5
Area of Circle = 78.5397

[Process completed - press Enter]
```

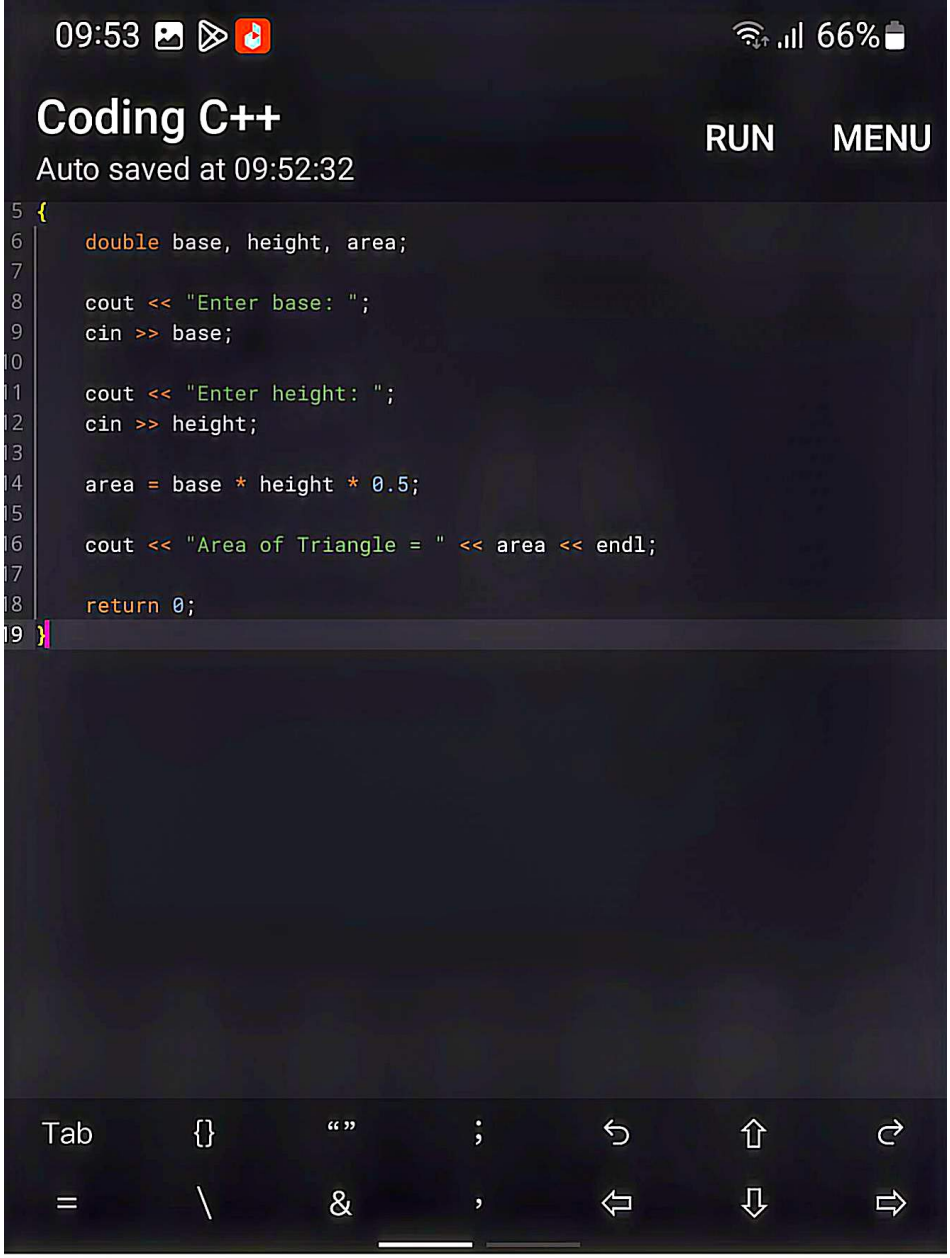
CamScanner


09:52
66%
Coding C++
Auto saved at 09:52:20
RUN
MENU
#include <iostream>
using namespace std;
int main()
{
double length, width, area;
cout << "Enter length: ";
cin >> length;
cout << "Enter width: ";
cin >> width;
area = length * width;
cout << "Area of Rectangle = " << area << endl;
return 0;
}
Tab
{}
“”
;
=
\
&
,

CamScanner


09:52
66%
Compile Result
Enter length: 10
Enter width: 5
Area of Rectangle = 50
[Process completed - press Enter]

CamScanner


09:53
66%
Coding C++
Auto saved at 09:52:32
RUN
MENU
{
double base, height, area;
cout << "Enter base: ";
cin >> base;
cout << "Enter height: ";
cin >> height;
area = base * height * 0.5;
cout << "Area of Triangle = " << area << endl;
return 0;
}
Tab
{}
“”
;
=
\
&
,

CamScanner

09:53 65%

Compile Result

```
Enter base: 10
Enter height: 6
Area of Triangle = 30

[Process completed - press Enter]
```

CamScanner

09:54 65%

Coding C++

Auto saved at 09:54:05

RUN MENU

```cpp
#include <iostream>
#include <iomanip>
using namespace std;

int main()
{
    double item1 = 12.95;
    double item2 = 24.95;
    double item3 = 6.95;
    double item4 = 14.95;
    double item5 = 3.95;

    double subtotal, tax, total;

    subtotal = item1 + item2 + item3 + item4 + item5;
    tax = subtotal * 0.06;
    total = subtotal + tax;

    cout << fixed << setprecision(2);

    cout << "Price of item 1 = $" << item1 << endl;
    cout << "Price of item 2 = $" << item2 << endl;
    cout << "Price of item 3 = $" << item3 << endl;
    cout << "Price of item 4 = $" << item4 << endl;
    cout << "Price of item 5 = $" << item5 << endl;

    cout << "Subtotal = $" << subtotal << endl;
    cout << "Sales Tax = $" << tax << endl;
    cout << "Total = $" << total << endl;

    return 0;
}
```

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09:54
65%
Compile Result
Price of item 1 = $12.95
Price of item 2 = $24.95
Price of item 3 = $6.95
Price of item 4 = $14.95
Price of item 5 = $3.95
Subtotal = $63.75
Sales Tax = $3.82
Total = $67.58
[Process completed - press Enter]

CamScanner